



API Design for Network Route Optimization

Add Node

****POST**** `/nodes`

****Request Body:****

```
```json
{
 "name": "ServerA"
}
```
```

****Response (201 Created):****

```
```json
{
 "id": 1,
 "name": "ServerA"
}
```
```

****Errors:****

* 400: Name missing or duplicate

Add Edge

****POST**** `/edges`



****Request Body:****

```
```json
{
 "source": "ServerA",
 "destination": "ServerB",
 "latency": 12.5
}
```
```

****Response (201 Created):****

```
```json
{
 "id": 1,
 "source": "ServerA",
 "destination": "ServerB",
 "latency": 12.5
}
```
```

****Errors:****

* 400: Source/destination missing, latency ≤ 0 , duplicate edge, nodes not found

3 Get Shortest Route

****POST**** `/routes/shortest`

****Request Body:****

```
```json
{
 "source": "ServerA",
 "destination": "ServerD"
}
```



```
}
...
```

**\*\*Response (200 OK, Path Exists):\*\***

```
```json  
{  
  "total_latency": 23.4,  
  "path": ["ServerA", "ServerB", "ServerD"]  
}  
...
```

****Response (404 Not Found, No Path):****

```
```json  
{
 "error": "No path exists between ServerA and ServerD"
}
...
```

**\*\*Errors:\*\***

\* 400: Invalid or non-existent nodes

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## ## 4 Get Route Query History

**\*\*GET\*\*** `/routes/history`

**\*\*Query Parameters (optional):\*\***

- \* `source` – filter by source node
- \* `destination` – filter by destination node
- \* `limit` – number of records
- \* `date\_from` / `date\_to` – filter by timestamp

**\*\*Response (200 OK):\*\***



```
```json
[
  {
    "id": 1,
    "source": "ServerA",
    "destination": "ServerD",
    "total_latency": 23.4,
    "path": ["ServerA", "ServerB", "ServerD"],
    "created_at": "2026-02-20T14:32:00Z"
  },
  {
    "id": 2,
    "source": "ServerB",
    "destination": "ServerC",
    "total_latency": 10.1,
    "path": ["ServerB", "ServerC"],
    "created_at": "2026-02-20T15:10:00Z"
  }
]
```
```

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### ## Optional (Nice-to-Have APIs)

```
* **GET** `/nodes` → List all nodes
* **GET** `/edges` → List all edges
* **DELETE** `/nodes/{id}` → Delete node
* **DELETE** `/edges/{id}` → Delete edge
```